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Where Minds Pollinate

Class 12
BIOLOGY
Chapter 7

NOTES
NBF book

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M. Imp

LQ+
SQS
MUST

Chapter # 07 : HUMAN SKELETAL SYSTEM

Significance :

- Support
- Locomotion
- Sustain against gravitational force.

Part 1 : Osteology

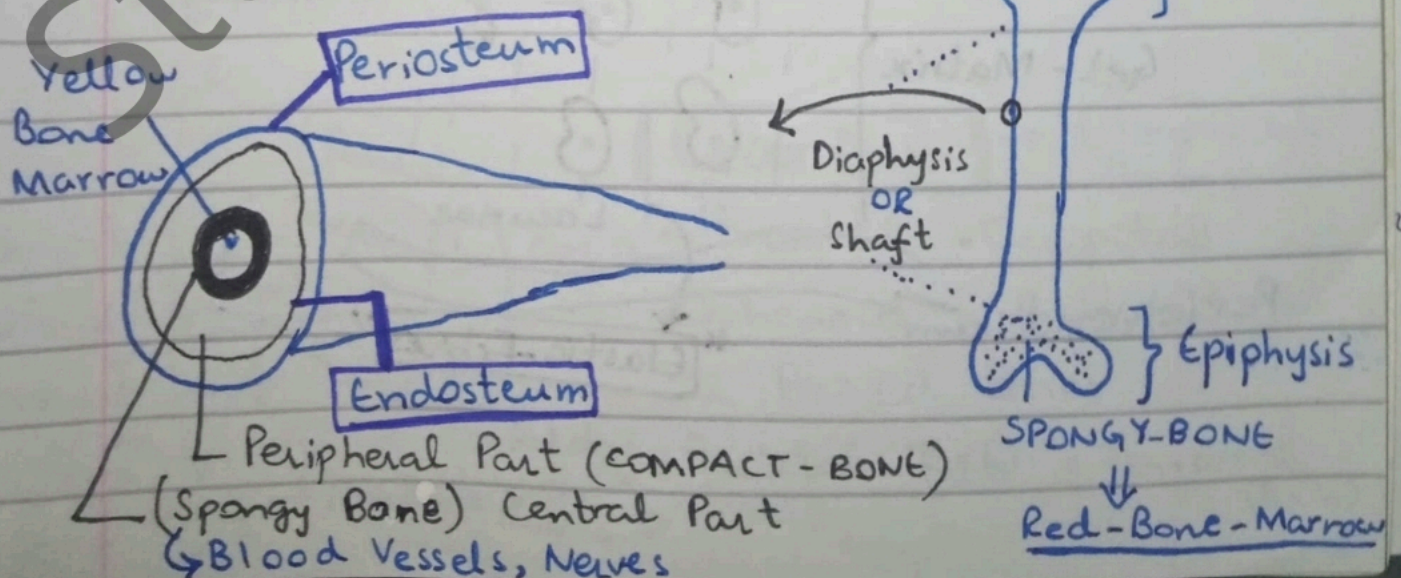
Study of bones and cartilage

Part 2 : Myology

Study of MUSCLES

Human Skeleton

a) Structure of Bone :



Osteogenic Stem Cells :

1) OSTEOBLAST

- Bone-forming cell
- Unmineralized substances production.

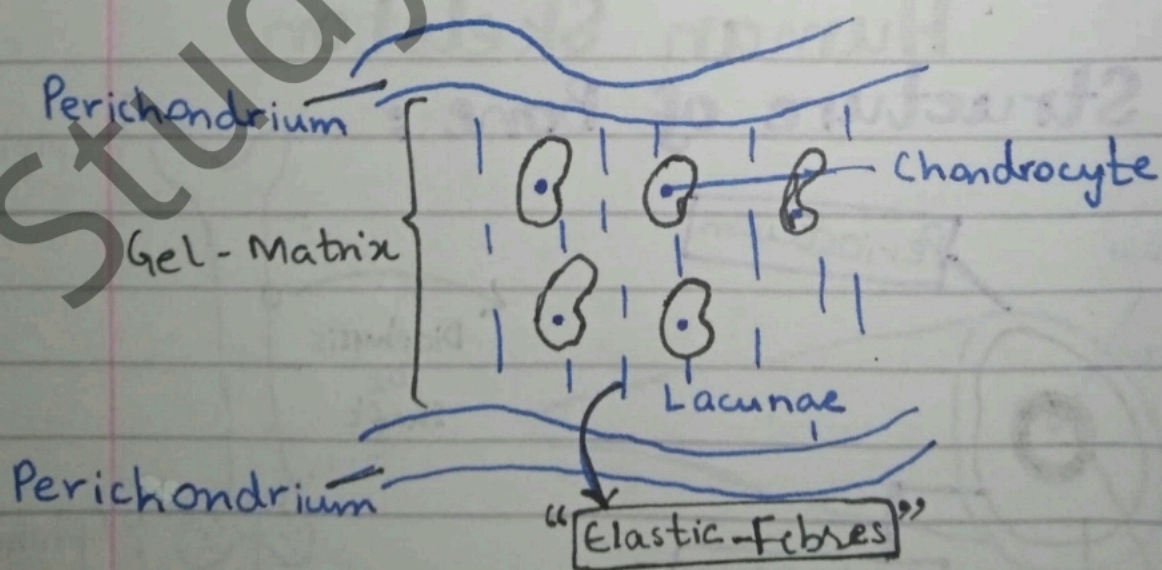
2) OSTEOCYTE

- Mature bone cell
- 'Mineral deposition' in bone.

3) OSTEOCLAST (Bone eating cell)

Resorption (Ca^{++} , Mg, P $\rightarrow$ Back into Blood)

b) Cartilage (SOFT CONNECTIVE TISSUE):



Types:

1) Hyaline Cartilage:

- At the end of 'LONG BONES'.
- Larynx, trachea, nose.

2) Fibrocartilage:

- Between two successive VERTEBRAE.

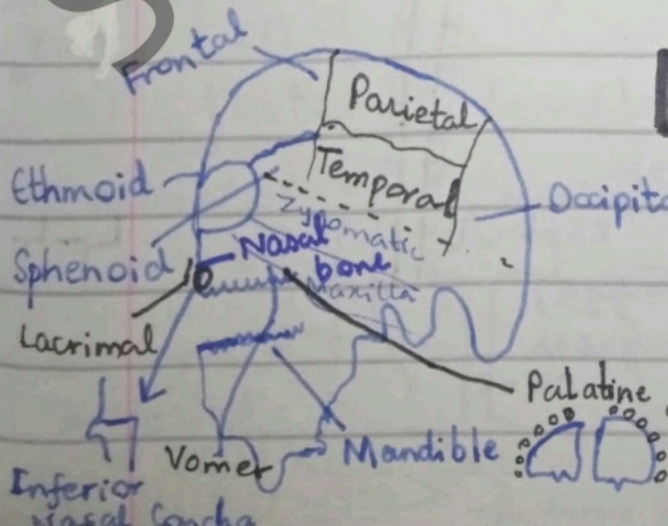
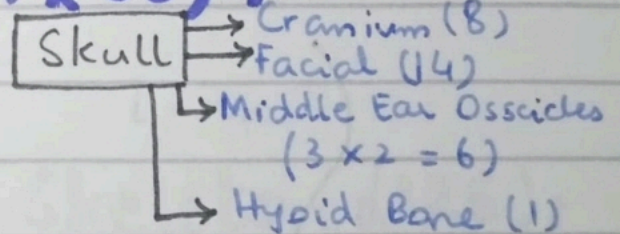
3) Elastic Cartilage:

- Pinna of Ear, Epiglottis.

Components of Human Skeleton

Axial Skeleton (80):

- 1) Skull (29)
- 2) Sternum (1)
- 3) Ribs (24)
- 4) Vertebral Column (26)



Cranium → 4 Unpaired Bones

- Frontal
- Occipital
- Sphenoid
- Ethmoid

2 Paired Bones

- Parietal (R x L)
- Temporal (R x L)

Facial Bones

6 - Paired

Lacrimal

Nasal Bone

Inferior Nasal Concha

Zygomatic → cheek bone

Maxilla → Upper Jaw

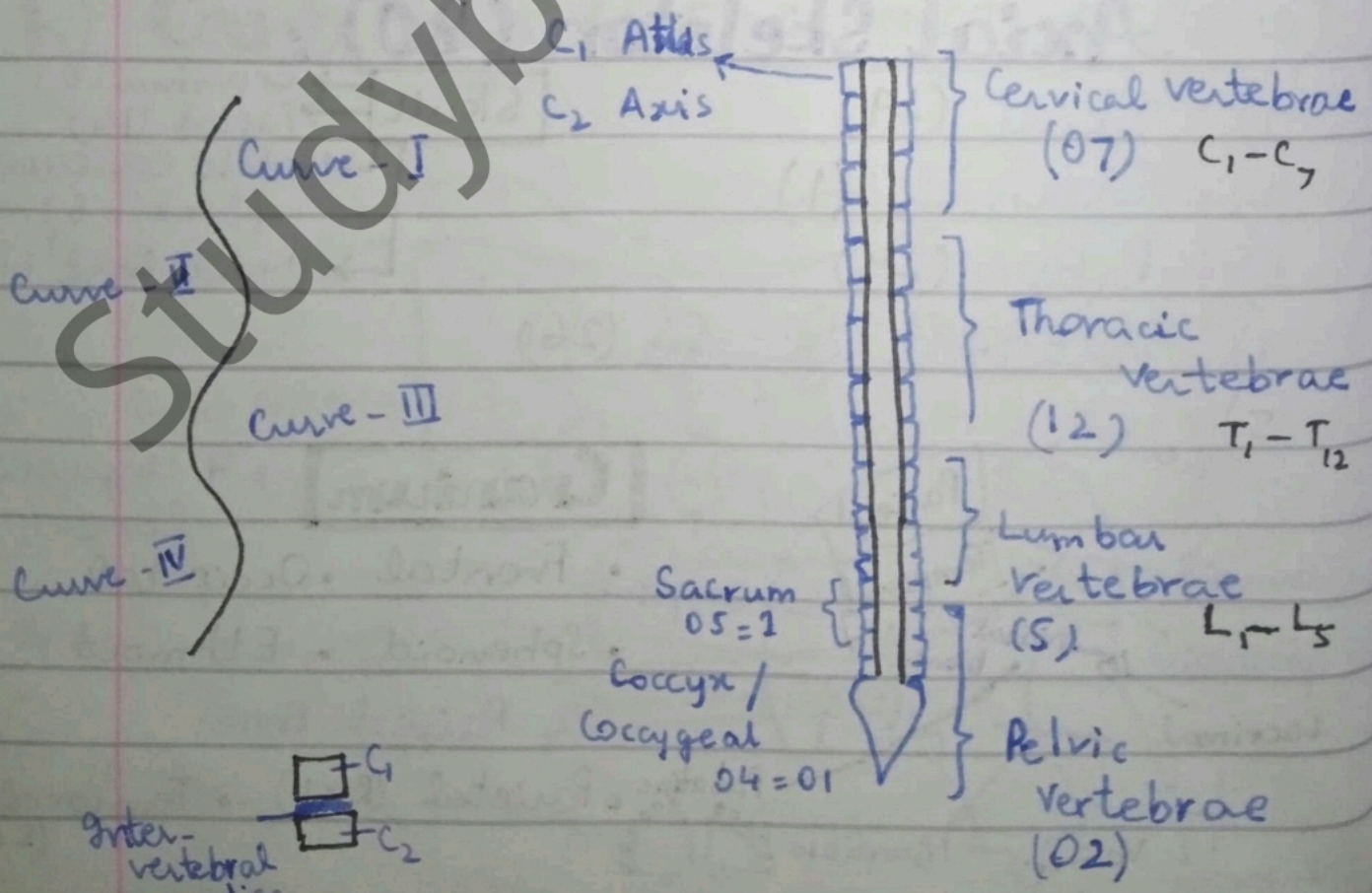
Palatine → Roof of the mouth (Right + Left)

2 - Unpaired

Mandible

Vomer

Vertebral Column :



Ribs :

• 12 → Right

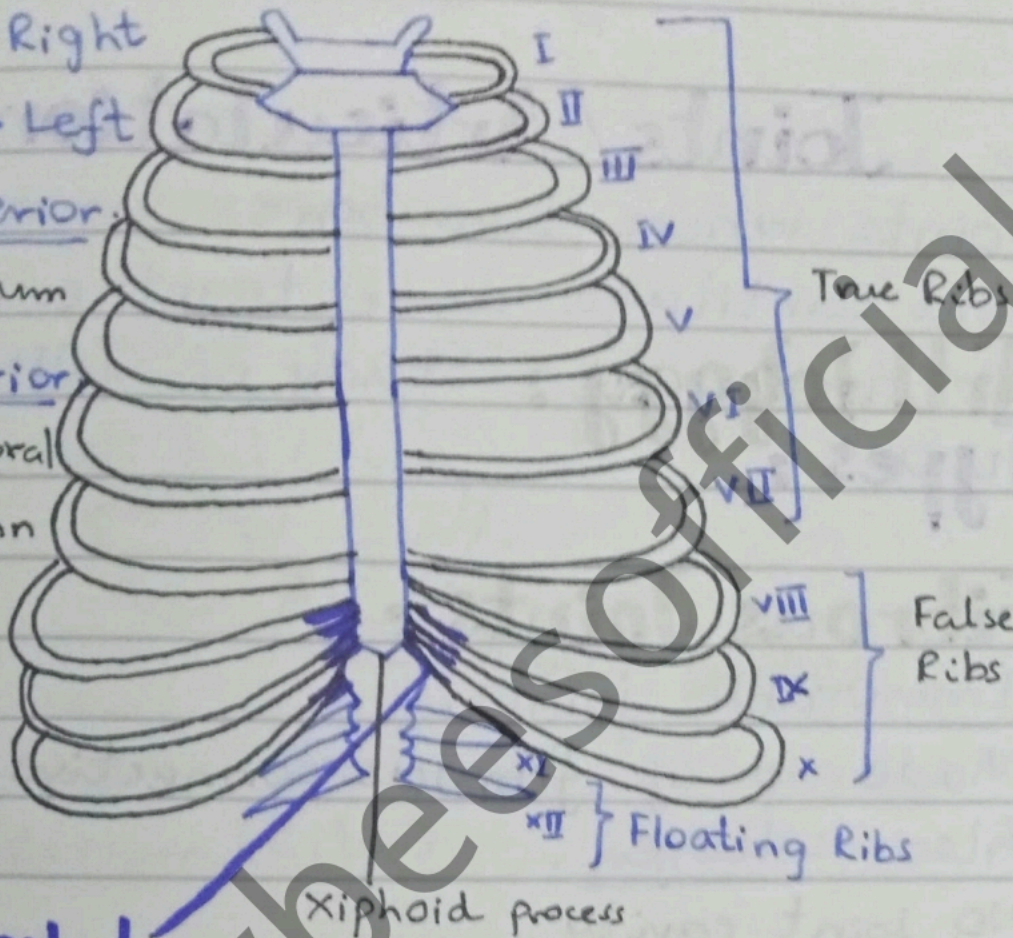
• 12 → Left

• Anterior

Sternum

• Posterior

Vertebral
Column



Costal
Cartilage

Appendicular Skeleton (126):

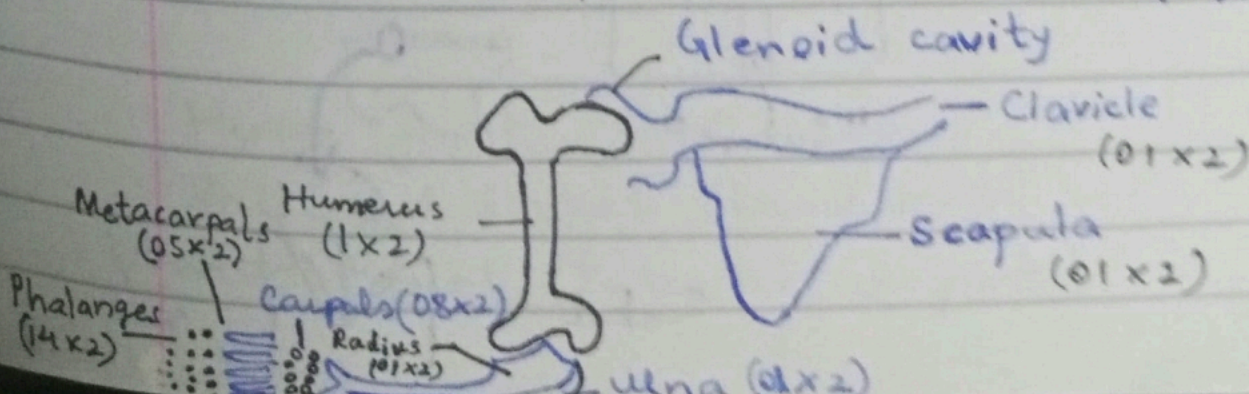
Present in Appendages

(4)

(60)

(64)

a) Pectoral Girdle + Forelimb (Upper-Limb)



b) Pelvic Girdle + Hind-limb (62)
(2) (60)

Joints/Articulation

"Points where two bones or a bone and cartilage come together"

Arthrology: Study of JOINTS.

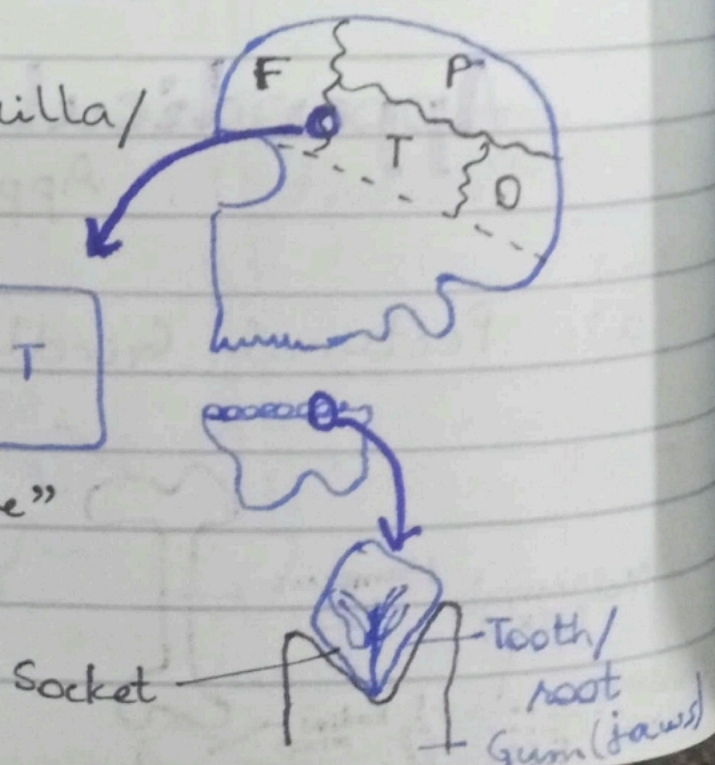
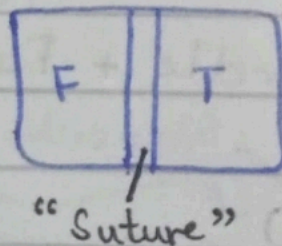
Types:

1) Fibrous Joints:

- Immovable joints.
- Made up of fibrous connective tissue
- Also collagen.
- NO joint cavity.

e.g. Sutures in SKULL

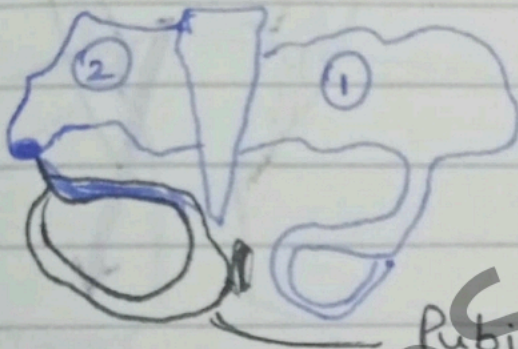
Tooth root in
socket of maxilla/
mandible.



2) Cartilagenous Joints :

Slightly movable joints.

Two bones in these joints are joined by cartilagenous.



Pubic symphysis

Costal cartilage attached to sternum (Ribs).

Hyaline cartilage

Fibrous cartilage

Elastic cartilage

3) Synovial Joints :

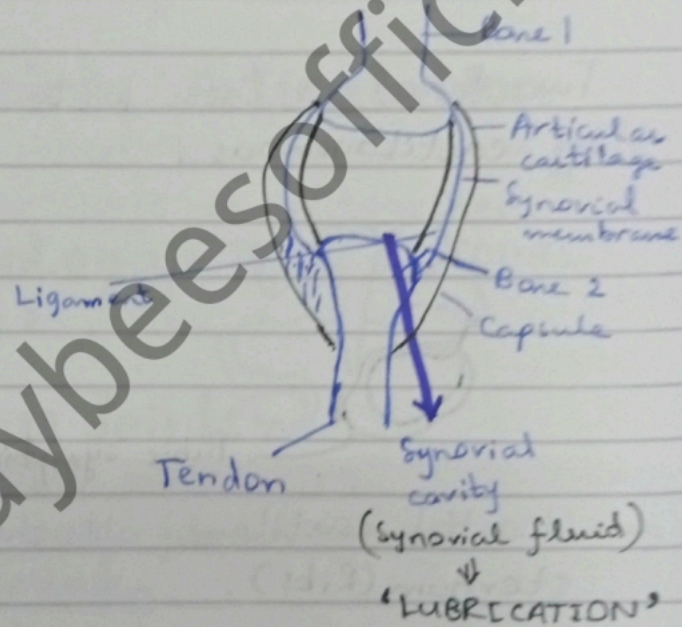
Freely movable joints.

Hinge joint (Wrist, Elbow)

Ball and socket (Shoulder)

Pivot joint (Intervertebral joints)

Gliding (Wrist, Ankle)



Bone Fracture

(Medical term used for broken bone)

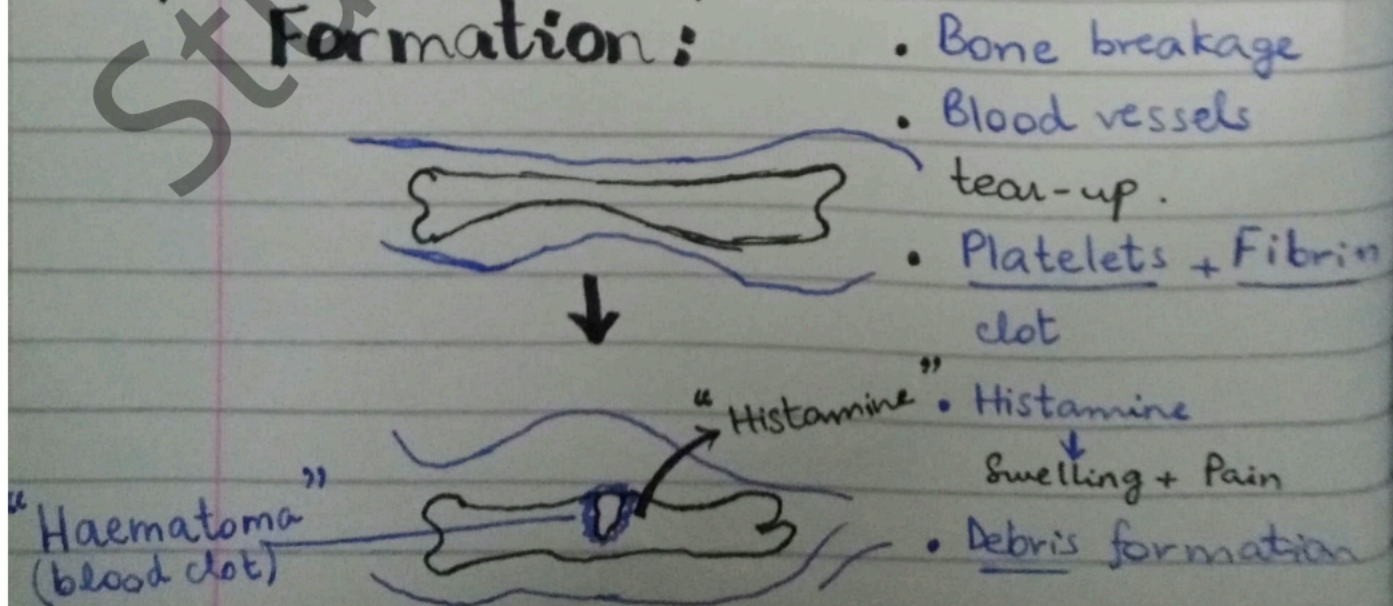
External Force > Bone strength

Types of Fracture:

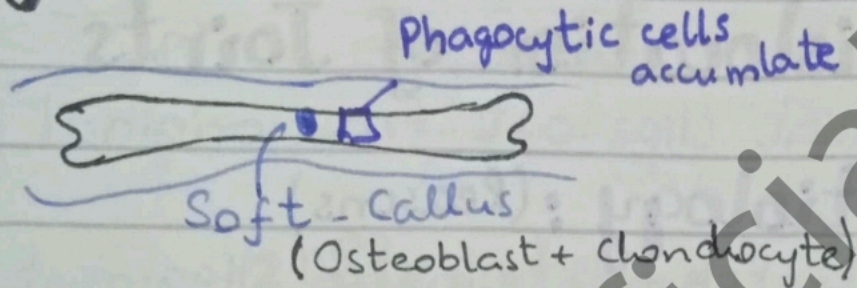
- Simple / Closed Fracture:**
Skin remains INTACT.
- Compound / Open Fracture:**
Bone ends penetrate skin.
- Complicated Fracture:**
If bone affects surrounding organs.

Bone Repair

1) Haematoma / Blood-Clot Formation:



2) Fibrocartilage Formation/ Soft - Callus Formation :

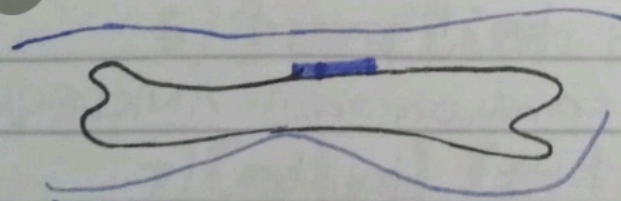


- Phagocytosis → clean up debris
- Periosteum → Osteoblast + chondrocytes
"Soft-Callus"
- Swelling "REDUCED"
- 3-4 weeks

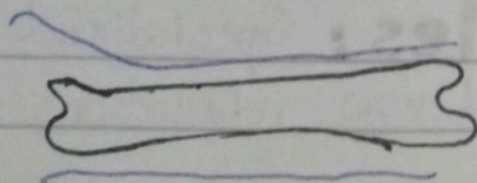
3) Bony - Callus Formation/ Hard Callus Formation :

Osteoblast → Osteocyte

Osteoclast → remove → Chondrocytes



- ## 4) Remodelling :
- Excessive bone cells will be removed by osteoclast and bone will regain its original look.



Injuries to Joints

1) Dislocation of Joints

Joint slips out from original position

Etiology: (Reasons)

- Stretch / tearing of ligament or tendon.
- Rheumatoid arthritis

Symptoms:

- Swelling of joints
- Inflammation

R_x: Reduced Reduction /
Closed Reduction

Surgery

2) Sprain

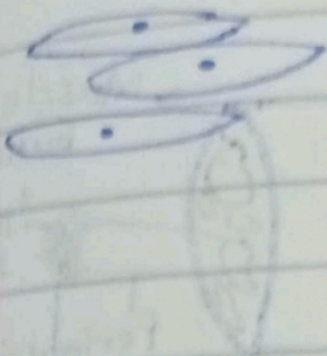
- Injury to ligament
- Most common in ANKLE, WRIST and KNEE joints.

- Ace wraps (Elastic-bandage)
↳ (to immobilize the effected area)

MYOLOGY → Study of **MUSCLES**

Muscles: Specialized tissue that can undergo relaxations and contraction.

Smooth



Non-striated muscles

Mononuclei

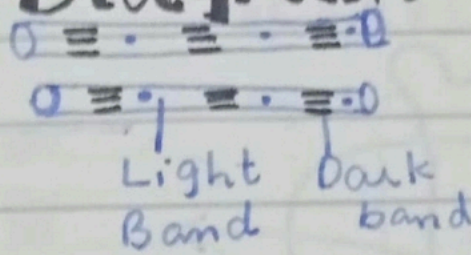
Involuntary

Spleen,
Liver

Secretions

Nervous
System

1) **Skeletal Diagram**



2) **Striations**

Striated muscles

3) **Nuclei**

Polynuclei

4) **Contractions**

Voluntary

5) **Presence Sites**

On Bones.

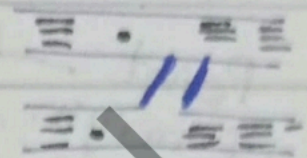
6) **Function**

Movement

7) **Control Centre**

Nervous
System

Cardiac



Striated muscles

Mononuclei

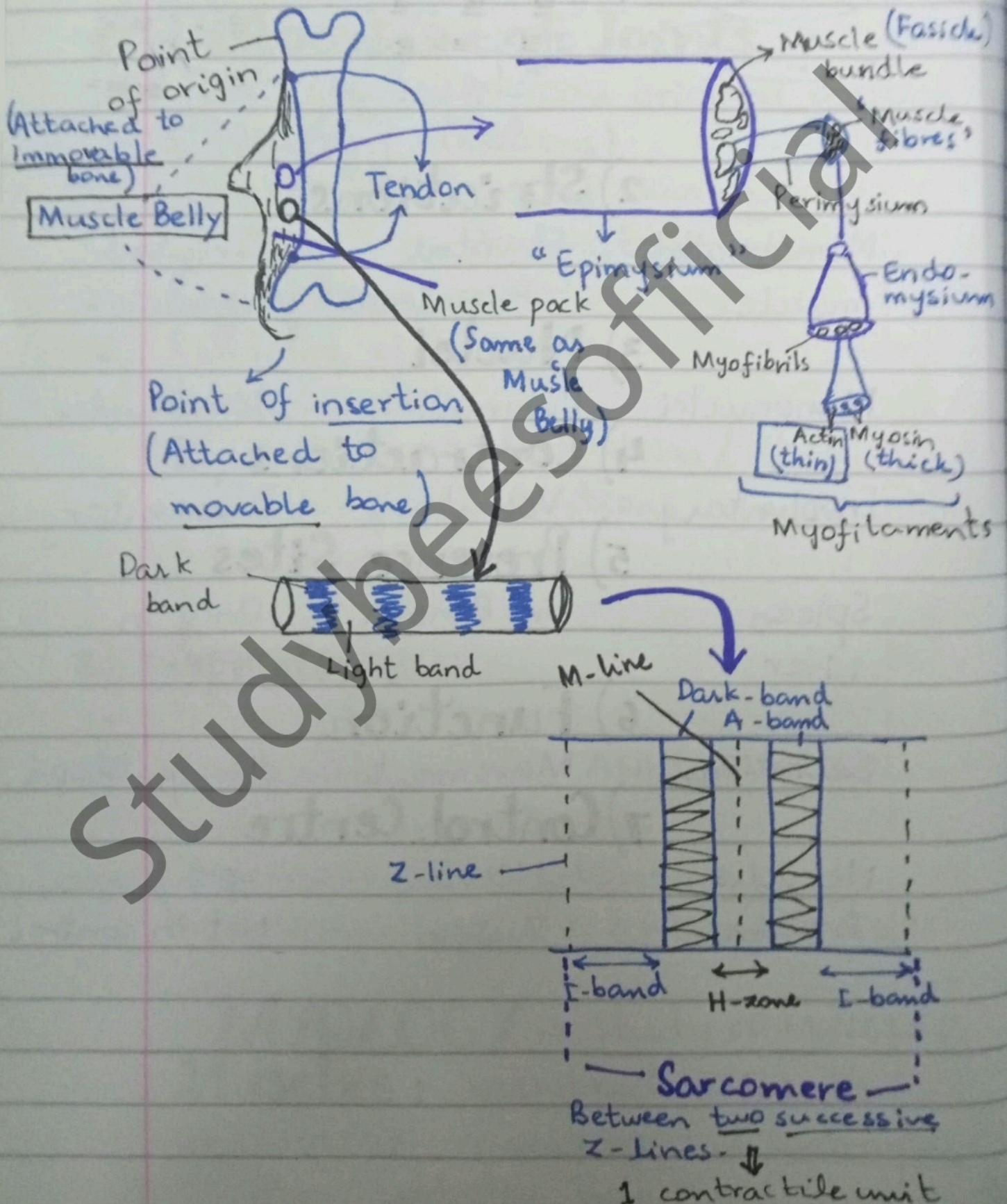
Involuntary

Only in
HEART.

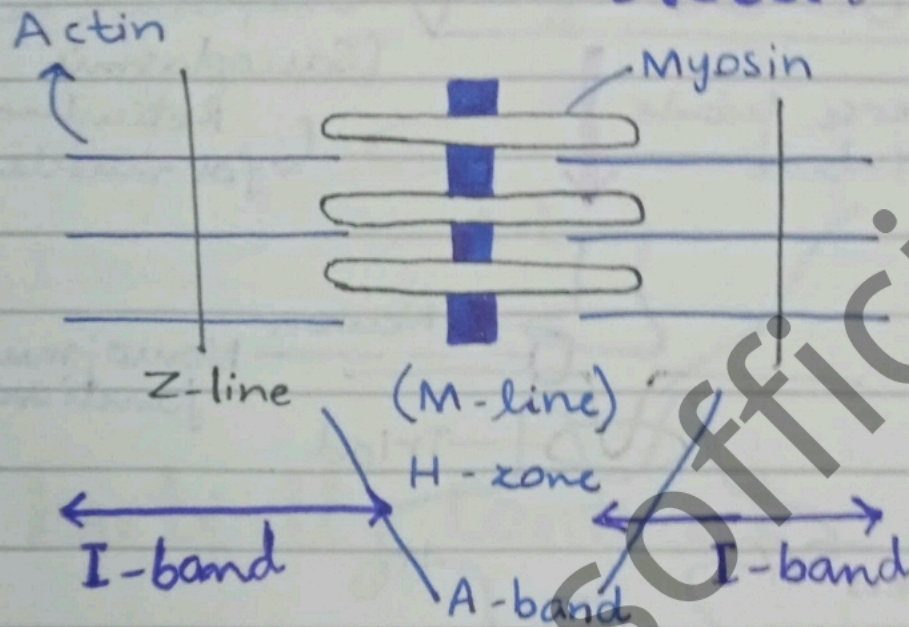
Pump-blood

Spontaneous/
Not in control

Structure of Skeletal Muscle



Arrangement of Myosin / Actin

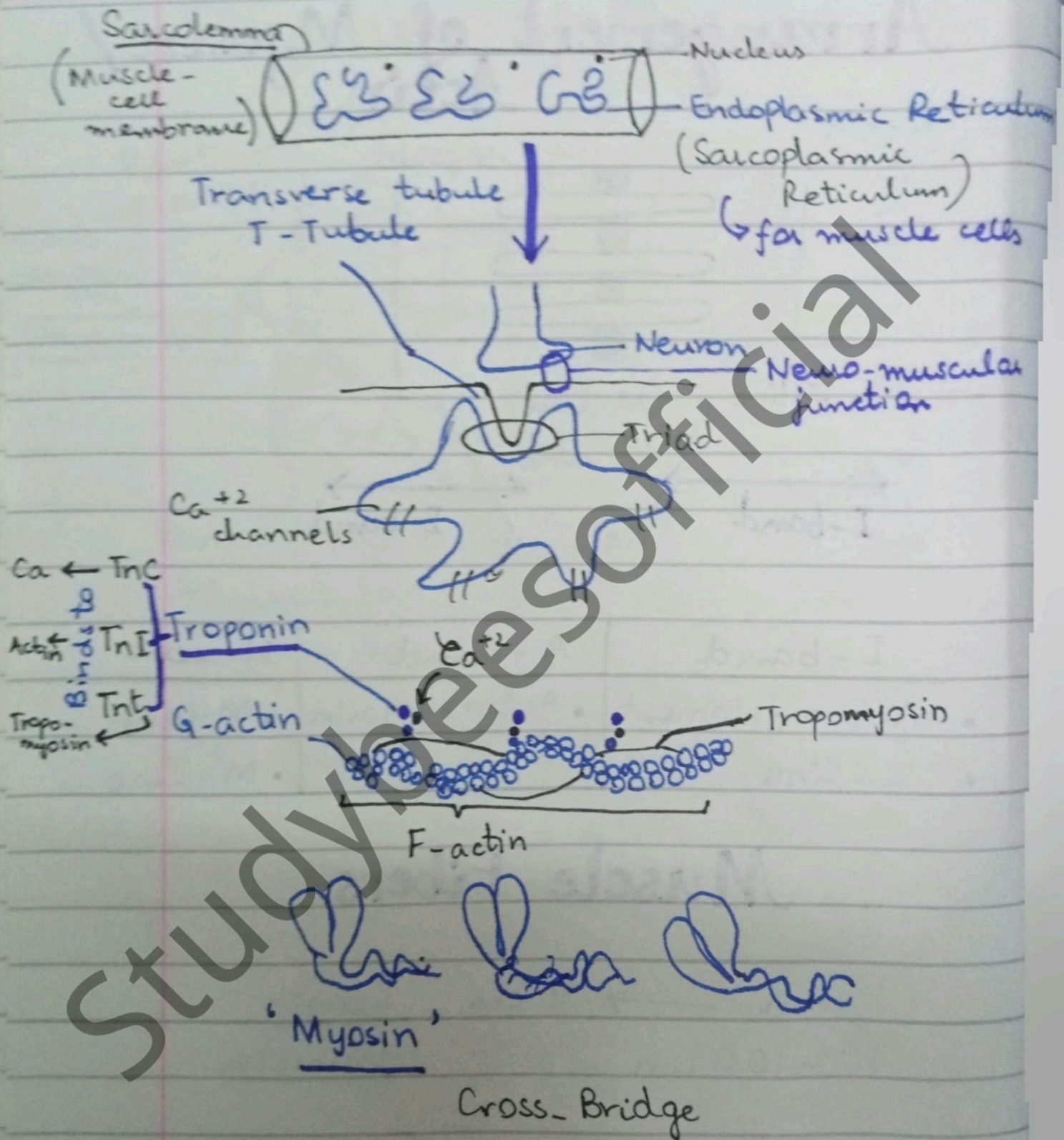


I - band	A - band	H - zone
• <u>Actin</u> filament • Z - line	• Actin + Myosin	• Myosin • M - line

Muscle Fibers

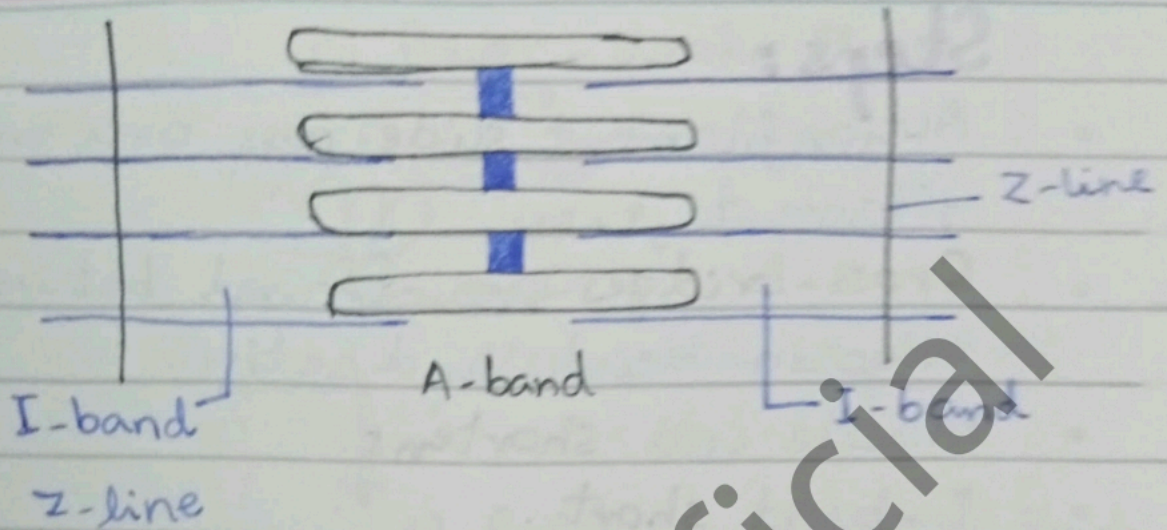
1 mm → 4 cm
10 - 100 μ m

Length
Diameter

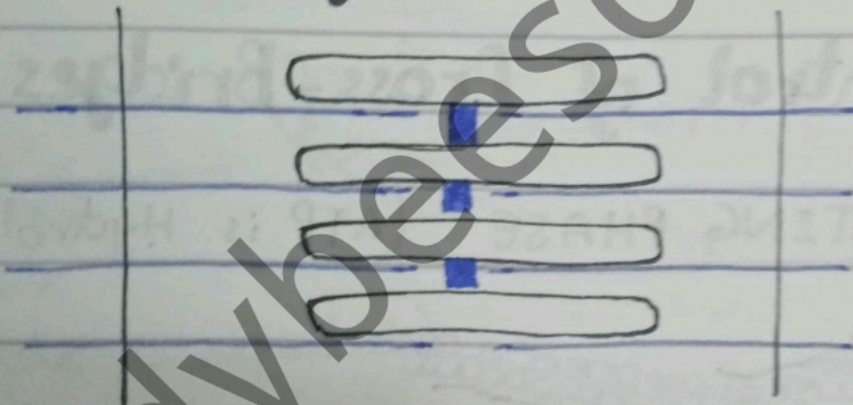


Sliding Filament Model

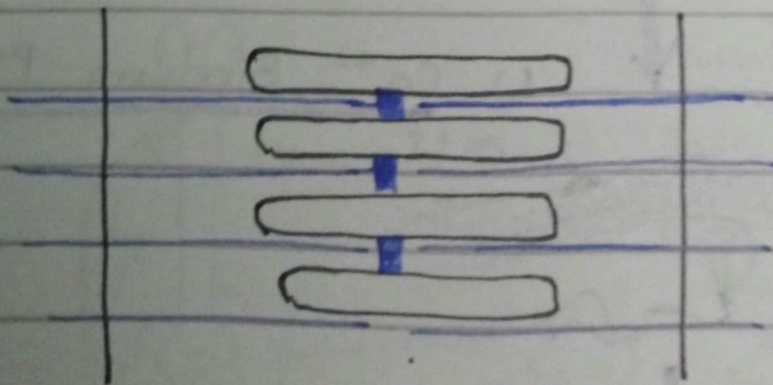
a) Muscle at Rest



b) Partially Contract



c) Fully Contracted Muscle

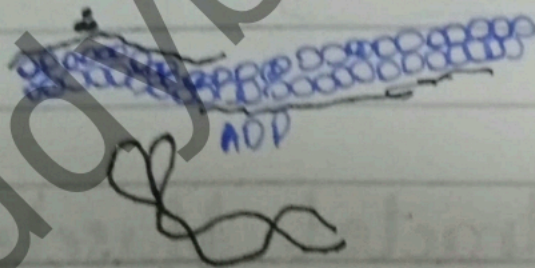


Steps:

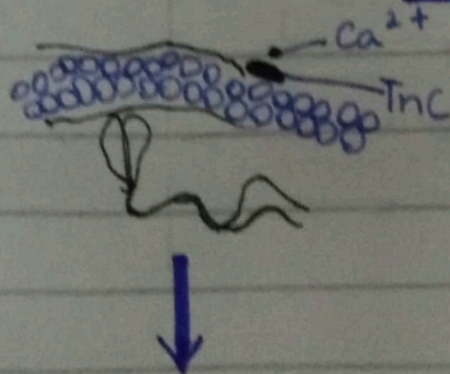
- Actin filament slide pass over myosin filament.
- Cross-bridges are formed between myosin head and actin.
- Muscle cell shortens.
- I-band short.
- Z-lines move closer to each other.
- H-zone disappears.

Control of Cross-Bridges

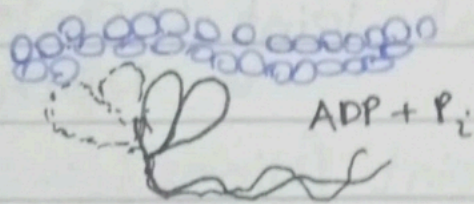
a) RESTING PHASE (ATP is Hydrolyzed)



b) Ca^{+2} - Binding Phase



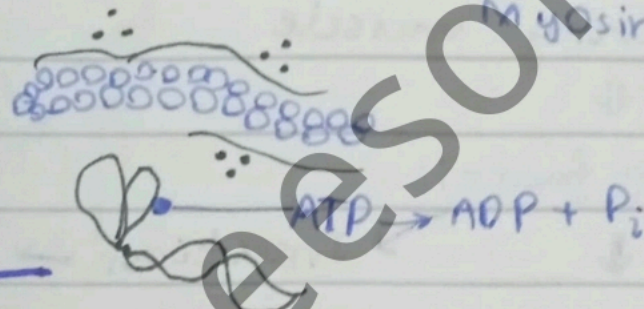
c) Power - Stroke Phase



Dissociation of $ADP + P_i$

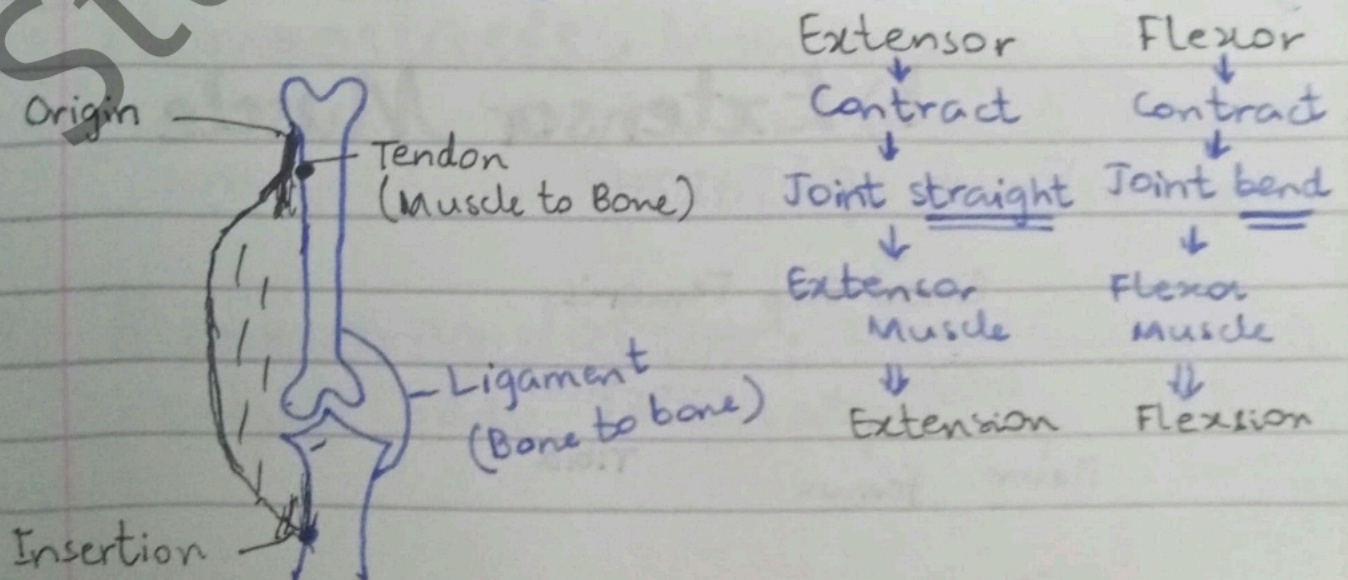


d) Breaking of Cross-Bridges (New ATP binds to Myosin)



Resting Phase
(cycle is repeated)

Antagonistic Action of Skeletal Muscles



Movement of Knee Joint

- Tibio-femoral joint (Located between TIBIA and FEMUR)
- Synovial joint (Hinge joint)

a) Flexor Muscle

Back of 'THIGH'

Hamstring muscle



Bicep femoris



Two origin

Top of Femur / Pelvic
Girdle

Insertion → Tendon



Two portion



Upper part of
Tibia and Fibula

b) Extensor Muscle

Front of 'THIGH'



Quadricep Femoris

origin

ileum

femur

insertion

Tibia

